

Project Development Phase

Preprocessing steps and business questions

Date	30 June 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	2 Marks

Data Preprocessing Steps

Step 1: Data Collection

- Imported the retail sales dataset into Tableau.
 - Verified that the dataset was successfully loaded.
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Step 2: Data Inspection

- Reviewed all columns and their data types.
 - Checked the total number of records and attributes.
 - Examined the dataset structure.
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Step 3: Handling Missing Values

- Checked for missing or null values.
 - Removed or handled incomplete records where necessary.
 - Ensured all important fields contained valid data.
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Step 4: Data Cleaning

- Removed duplicate records.
 - Corrected inconsistent values.
 - Standardized category names and text formatting.
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Step 5: Data Type Validation

- Converted Sales Volume and Price to numeric data types.

- Ensured categorical fields such as Product Category, Product Position, Promotion, and Season were correctly identified.
 - Verified date fields (if applicable).
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Step 6: Feature Preparation

- Renamed fields for better readability.
 - Organized dimensions and measures.
 - Created hierarchies where required.
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Step 7: Calculated Fields

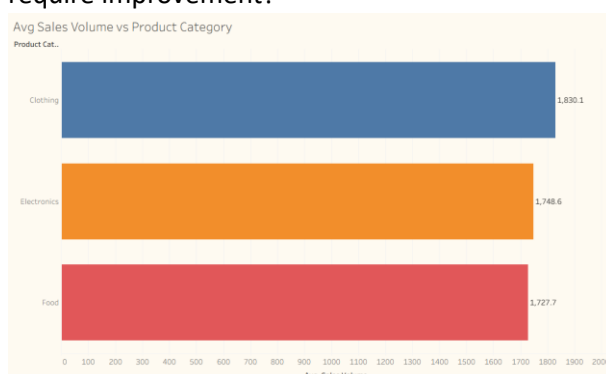
- Created Average Sales Volume.
 - Calculated Average Price.
 - Used aggregation functions for business analysis.
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Step 8: Data Validation

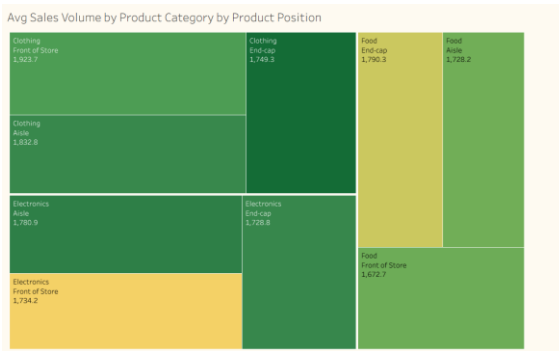
- Cross-checked totals and averages.
- Verified charts against source data.
- Ensured data consistency before visualization.

Business Questions

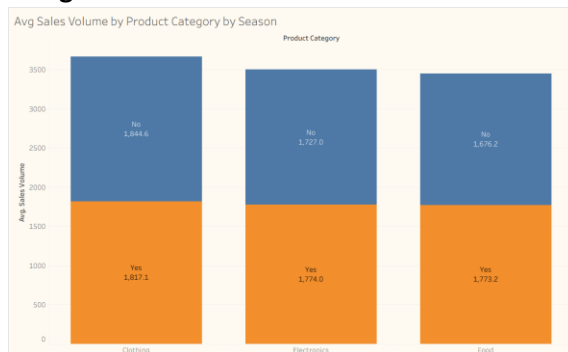
1. Which product categories generate the highest average sales volume, and which categories require improvement?



2. How does product placement influence the average sales volume of different product categories?



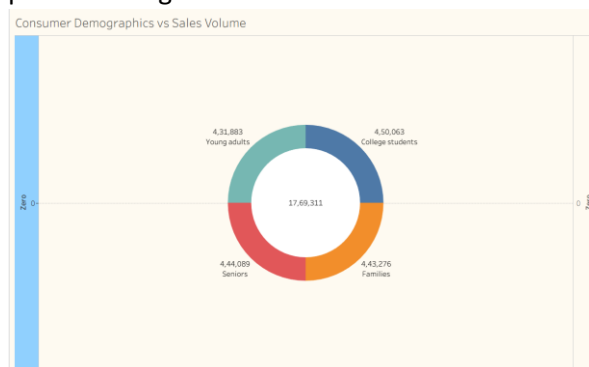
3. How do seasonal changes affect the average sales volume across different product categories?



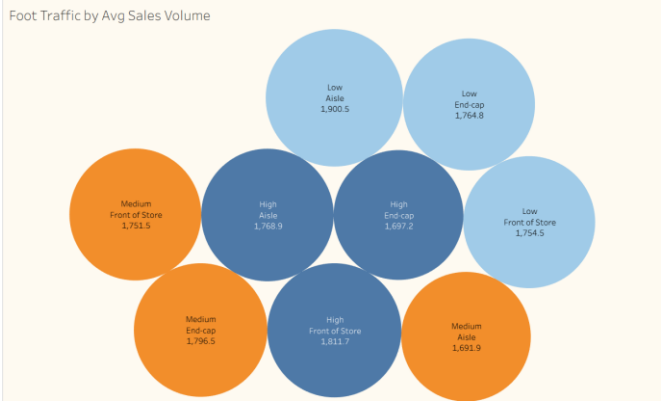
4. How do our product prices compare with competitor prices, and how might pricing influence sales performance?



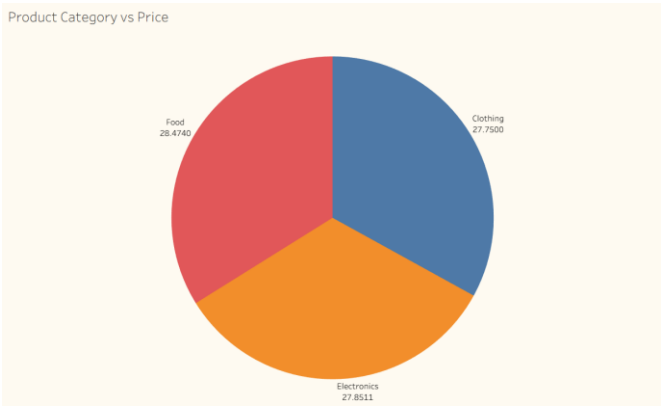
5. Which customer demographic groups contribute the highest sales volume for different product categories?



6. What is the relationship between store foot traffic and average sales volume?



7. How does product pricing vary across different product categories, and what impact could this have on customer purchasing decisions?



8. How do promotional activities influence product price and average sales volume across different product categories?

Promotion of Product Category on Price and Sales			
Promotion	Product Ca..	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
	Electronics	27	1,726
	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782